AeroFlow – Flight Disruption Recovery & Passenger Service Platform

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Under Guidance

of

Internal Guide

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Changa

Acknowledgement

Knowledge in itself is a continuous process. At this moment of our substantial enhancement, We rarely find words to express our gratitude towards those who were constantly involved with us.

The completion of any inter disciplinary project depends upon coordination, cooperation and combined efforts of several resources of knowledge, creativity, skill, energy and time. The work being accomplished now, we feel our sincerest urge to recall and knowledge through these lines, trying our best to give full credit wherever it deserves.

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It's our good fortune that we had support and well wishes of many. We are thankful to all and those names which have been forgotten to acknowledge here but contributions have not gone unnoticed.

With Sincere Regards,

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Project Profile

❖ Project Profile

Project Name: AeroFlow – Flight Disruption Recovery & Passenger Service Platform

Type of Application: Web-based flight management and passenger service application

Project Description: AeroFlow is a web application designed to support flight-related operations and passenger services, particularly during flight disruptions. The project aims to organize flight and airport information and support recovery-related workflows. The system provides a platform for managing relevant information and improving the handling of flight disruptions.

Team Size: 2

Front End: HTML , CSS , Java Script, Django Templates

Back End: Python, Django Framework

Tools used:

- **Programming Language:** Python
- **Web Framework:** Django (4.2 or later, below 5.1)
- **Database:** SQLite
- **PDF Generation:** ReportLab
- **QR Code Generation:** qrcode
- **Image Processing:** Pillow
- **Code Editor:** Visual Studio Code
- **Version Control:** Git and GitHub
- **Browser:** Google Chrome or Microsoft Edge

Introduction to tools

❖ Introduction to Tools

➢ Front End Tool:

HTML (HyperText Markup Language)

HTML is used to create the basic structure of web pages. It defines elements such as headings, forms, buttons, tables, links, and other components displayed to users.

In AeroFlow, HTML is used to design pages for passenger registration, login, flight search, booking details, and other application features.

CSS (Cascading Style Sheets)

CSS is used to control the appearance and layout of web pages. It helps define colors, fonts, spacing, borders, alignment, and page layouts.

In AeroFlow, CSS is used to make the application interface clear, consistent, and user-friendly.

JavaScript

JavaScript is a scripting language used to add interactive behavior to web pages. It can respond to user actions, validate inputs, and update parts of the interface dynamically.

In AeroFlow, JavaScript can support interactive elements such as form validation, dropdown selections, and other client-side operations where implemented.

Django Template Language (DTL)

Django Template Language is used to display dynamic data in HTML templates. It supports template variables, loops, conditional statements, and template inheritance.

In AeroFlow, Django templates connect the user interface with data and information supplied by the Django back end.

➢ Back End Tool:

Python

Python is a high-level programming language known for its simple syntax and readability. It is widely used for web development, automation, data processing, and application development.

In AeroFlow, Python is used to implement application logic, process passenger requests, manage flight-related operations, and support booking and recovery workflows.

Django Framework

Django is a high-level Python web framework that supports the development of secure and maintainable web applications. It provides built-in features for URL routing, database operations, authentication, form handling, and template rendering.

In AeroFlow, Django is used to develop the main application, manage passenger and administrator operations, process bookings, and handle flight disruption recovery functions.

➢ Database Tool

SQLite

SQLite is a lightweight relational database management system that stores data in a database file. It does not require a separate database server and is suitable for small applications and prototypes.

In AeroFlow, SQLite is used to store and manage application data, including flight details, passenger information, bookings, and other records defined in the project.

System Study

1. Existing System

In the existing system, flight management and passenger services may involve separate processes for maintaining flight information, handling bookings, and communicating flight-related updates. During flight disruptions, passengers may face difficulties obtaining updated information and understanding the available recovery options.

Manual handling of flight records and passenger requests can require additional time and effort. Maintaining booking information, issuing travel documents, and coordinating recovery-related activities may also become challenging when information is not managed through a centralized platform.

The existing approach highlights the need for a web-based system that organizes flight information and supports passenger services more efficiently.

Limitations of the Existing System:

- Flight and passenger information may be maintained across different records or processes.
- Updating flight information and communicating changes may require additional effort.
- Managing passenger bookings and cancellations can be time-consuming.
- Handling disruption-related passenger requests may involve multiple steps.
- Generating and maintaining booking documents manually can be inconvenient.
- Tracking recovery-related activities may be difficult without a centralized system.

2. Proposed System

The proposed system, AeroFlow, is a web-based flight management and passenger service platform developed using Python and the Django framework. It provides a centralized application for managing flight information, passenger bookings, and selected services related to flight disruptions.

The system supports administrative operations and passenger-facing features through a web interface. It helps organize flight and airport records, manage booking-related activities, and support workflows such as cancellations, rescheduling requests, and disruption recovery.

The application uses a database to maintain relevant information and provides supporting functionality for generating PDF documents and QR codes where required.

Advantages of the Proposed System:

- Centralized management of flight, airport, and booking information.
- Convenient access to passenger services through a web interface.
- Support for passenger registration, login, and booking management.
- Easier handling of booking cancellation and rescheduling workflows.
- Generation of PDF booking documents and QR codes where implemented.
- Improved organization of flight disruption recovery activities.
- Reduced dependence on separate manual records.
- Better accessibility and maintainability through a web-based application.

3. Scope of the Proposed System

The scope of AeroFlow is to develop a web-based platform that supports flight management, passenger services, and disruption-related workflows. The application is designed to provide useful functions for both administrators and passengers.

The major areas covered by the system are as follows:

1. Flight Management: Managing flight details and updating flight status.
2. Airport Management: Maintaining airport information, including terminals and gates.
3. Passenger Management: Supporting passenger registration, login, and account-related operations.
4. Booking Management: Searching for flights, making bookings, viewing booking details, and managing reservations.
5. Cancellation and Rescheduling: Supporting cancellation and rescheduling requests according to the application's implemented workflows.
6. Boarding Pass Management: Generating boarding-pass information and associated QR codes when the required conditions are met.
7. Disruption Recovery: Supporting recovery-related processes, including applicable alternative-flight and passenger-service workflows.
8. Notifications: Sending relevant email notifications for supported booking and flight-related events.
9. Document Generation: Generating PDF documents, such as booking receipts, where implemented.
10. Administrative Operations: Providing authorized administrators with functions to manage relevant application records and requests.

The project focuses on organizing flight-related information and improving the management of passenger services through a centralized web application. It does not claim to replace the airline's actual operational control systems or guarantee integration with external airline databases.

4. Aim and Objectives of the Proposed System

4.1 Aim

The main aim of AeroFlow is to develop a web-based flight management and passenger service platform that organizes flight information, simplifies booking-related operations, and supports passenger service and recovery workflows during flight disruptions.

4.2 Objectives

The major objectives of the proposed system are:

1. To develop a centralized platform for managing flight and airport information.
2. To provide a user-friendly interface for passenger registration and authentication.
3. To enable passengers to search for available flights and manage their bookings.
4. To support booking cancellation and rescheduling request workflows.
5. To provide administrators with facilities to manage flights and update flight statuses.
6. To support boarding-pass generation with QR codes according to the defined workflow.
7. To organize disruption recovery processes and applicable passenger services.
8. To provide email notifications for relevant booking and flight-related events.
9. To generate PDF documents for supported booking operations.
10. To maintain application information in a structured database.
11. To improve the organization, accessibility, and efficiency of flight-related passenger services.

5. Feasibility Study

A feasibility study evaluates whether a proposed system can be developed and used successfully. It examines the technical resources, operational requirements, and financial considerations involved in developing and maintaining the application.

For AeroFlow, the feasibility study is divided into three categories: operational feasibility, technical feasibility, and economical feasibility.

5.1 Operational Feasibility

Operational feasibility determines whether the proposed system can be used effectively by its intended users and whether it supports their requirements.

AeroFlow provides a web-based interface through which passengers can access supported booking services and administrators can manage relevant flight and airport information. The system organizes information in a centralized application, making it easier for authorized users to perform routine operations.

The interface is designed to support common tasks such as logging in, searching for flights, viewing bookings, and processing supported requests. Users with basic computer and web-browser knowledge can operate the application with appropriate instructions.

Conclusion: AeroFlow is operationally feasible as an academic prototype because it provides a structured interface for managing flight-related information and passenger services.

5.2 Technical Feasibility

Technical feasibility evaluates whether the technologies, software, and development resources required for the project are available and suitable.

AeroFlow is developed using Python and the Django web framework. HTML, CSS, JavaScript, and Django templates can be used to build the user interface. SQLite is used for database management, while additional Python libraries support PDF generation, QR code creation, image processing, and configuration management.

The main technologies used are:

Technology	**Purpose**
Python	Application programming and business logic
Django	Web application development
HTML, CSS, JavaScript	User interface development
Django Template Language	Displaying dynamic information
SQLite	Database management
ReportLab	PDF document generation
qrcode	QR code generation
Pillow	Image processing
python-decouple	Configuration management
django-widget-tweaks	Django form customization

These technologies are suitable for developing and testing a web-based academic project. They provide the necessary functionality without requiring a complex infrastructure for the prototype.

Conclusion: AeroFlow is technically feasible because it uses established technologies and libraries that support its core application requirements.

5.3 Economical Feasibility

Economical feasibility evaluates the estimated cost of developing, operating, and maintaining the system.

AeroFlow uses Python, Django, SQLite, and the supporting libraries listed in its requirements file. These technologies are available as open-source software, which helps reduce software licensing expenses.

The project can be developed and tested on an existing computer using a compatible development environment and web browser. Therefore, the initial development cost can be kept relatively low.

The main potential expenses include:

- Computer hardware and electricity.
- Internet connectivity.
- Optional web hosting or cloud deployment.
- Any paid third-party services used during deployment or operation.

For an academic prototype, the project can be developed using existing resources without requiring substantial additional investment.

Conclusion: AeroFlow is economically feasible as an academic project because it relies primarily on open-source software and can be developed using commonly available computing resources.

System Analysis

4.1 Requirements Specification

This section explains what is needed to run and use AeroFlow.

A. Hardware requirements

You can document the following as the development environment requirements, adjusting them to match your computer.

- Computer or laptop
- Processor: Intel or AMD processor
- RAM: 4 GB minimum; 8 GB recommended
- Storage: Sufficient free disk space for the project and database
- Internet connection for development resources and email services, where required

B. Software requirements

Software	Purpose
Operating System	Windows or another compatible OS
Programming Language	Python
Web Framework	Django
Front-End Technologies	HTML, CSS, JavaScript and Django templates
Database	SQLite
Code Editor	Visual Studio Code, if used
Web Browser	Chrome or Edge, if used

C. Functional requirements

These describe what AeroFlow should do.

1. Allow passengers to register, log in and log out.
2. Allow passengers to search for flights.
3. Support flight booking and viewing booking details.
4. Allow passengers to submit cancellation or rescheduling requests.
5. Allow administrators to manage flight information.
6. Support airport, terminal and gate management.
7. Support flight-status updates and relevant notifications.
8. Generate booking receipts and eligible boarding passes.
9. Support applicable vouchers and disruption-recovery workflows.

D. Non-functional requirements

These describe the qualities expected of the application.

- Usability: The interface should be understandable and easy to navigate.
- Security: Authentication and access control should protect restricted operations.
- Reliability: The application should process and store data correctly.
- Maintainability: The code should be organized for easier updates.
- Performance: Common operations should complete within a reasonable time.
- Data integrity: Booking and seat information should follow the defined validation rules.

4.2 System Modules

Explain the main modules of AeroFlow in separate short paragraphs.

1. User Authentication Module

Handles passenger registration, login, session management and logout.

2. Flight Management Module

Allows authorized administrators to add, edit and manage flight details and status.

3. Booking Management Module

Supports flight reservations, passenger details, seat allocation and booking information.

4. Recovery Management Module

Supports applicable cancellation, rescheduling and alternative-flight workflows.

5. Airport Management Module

Manages airport information, terminals and gates.

6. Document and Notification Module

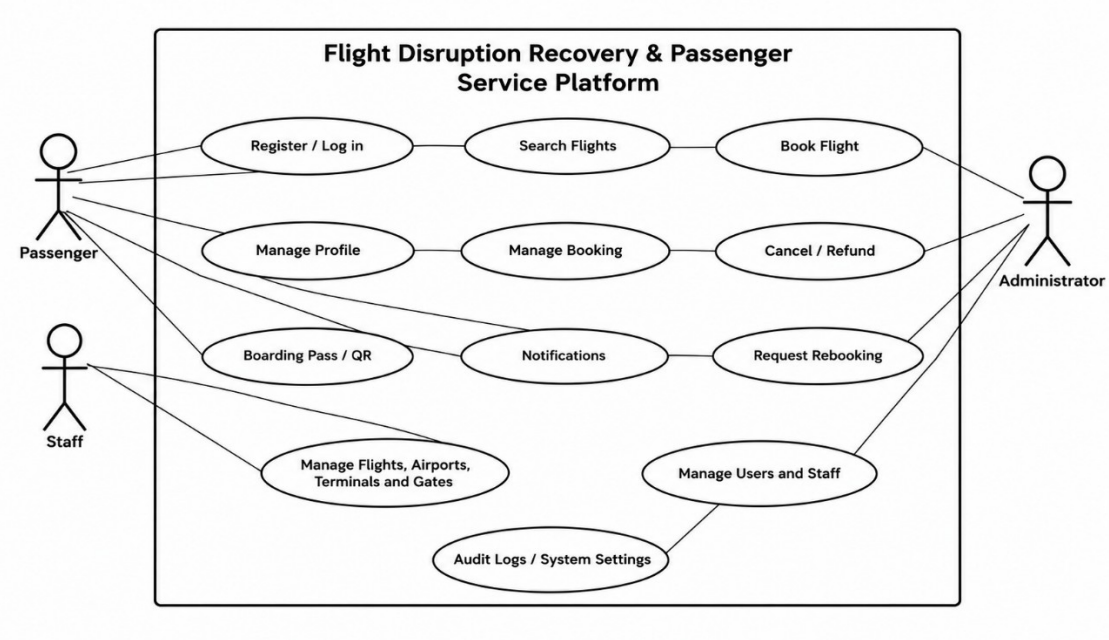
Supports booking receipts, boarding-pass generation, QR codes and relevant email notifications.

7. Voucher Management Module

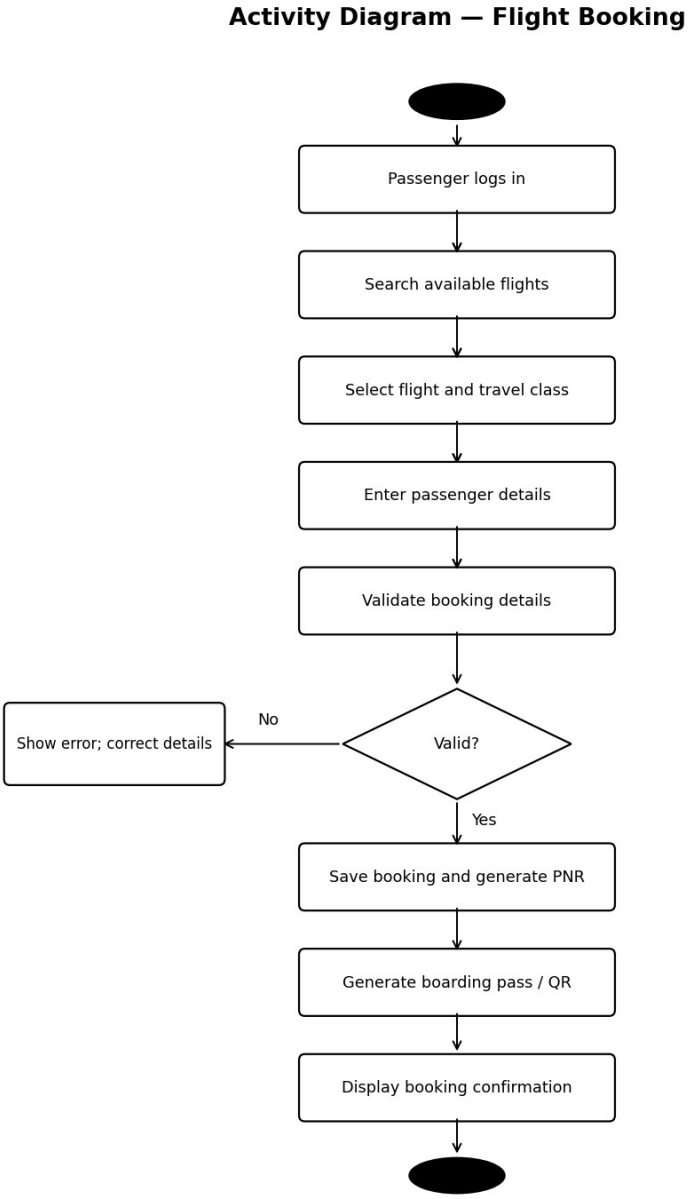
Manages applicable passenger vouchers and their use according to the implemented rules.

Only include modules that are present in your actual implementation.

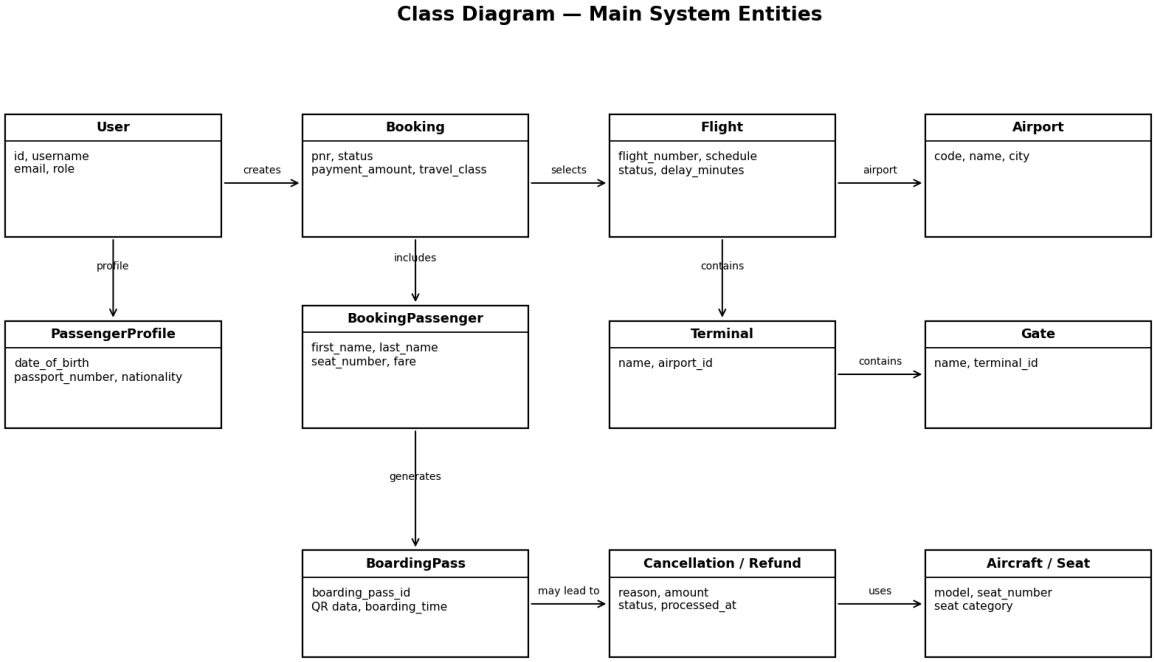
4.3 Use Case Diagram



4.4 Activity Diagram



4.5 Class Diagram



System Design

Data Dictionary

Accounts Passenger profile

Field Name	Data Type
id	INTEGER (Primary Key)
date_of_birth	DATE (Nullable)
passport_number	VARCHAR(30)
nationality	VARCHAR(60)
address	VARCHAR(255)
frequent_flyer_number	VARCHAR(30) (Nullable, Unique)
created_at	DATETIME
updated_at	DATETIME
user_id	BIGINT (Foreign Key, Unique)

Accounts Staff profile

Field Name	Data Type
id	INTEGER (Primary Key)
employee_id	VARCHAR(20) (Unique)
department	VARCHAR(30)
designation	VARCHAR(60)
created_at	DATETIME
updated_at	DATETIME
user_id	BIGINT (Foreign Key, Unique)

Accounts User

Field Name	Data Type
id	INTEGER (Primary Key)
password	VARCHAR(128)
last_login	DATETIME (Nullable)
is_superuser	BOOLEAN
username	VARCHAR(150) (Unique)
first_name	VARCHAR(150)
last_name	VARCHAR(150)
email	VARCHAR(254)
is_staff	BOOLEAN
is_active	BOOLEAN
date_joined	DATETIME

role	VARCHAR(20)
phone_number	VARCHAR(20)
created_at	DATETIME
updated_at	DATETIME

Accounts User Groups

Field Name	Data Type
id	INTEGER (Primary Key)
user_id	BIGINT (Foreign Key)
group_id	INTEGER (Foreign Key)

Accounts User User Permissions

Field Name	Data Type
id	INTEGER (Primary Key)
user_id	BIGINT (Foreign Key)
permission_id	INTEGER (Foreign Key)

Bookings Boarding pass

Field Name	Data Type
id	INTEGER (Primary Key)
boarding_pass_id	VARCHAR(16) (Unique)
boarding_time	TIME (Nullable)
pdf_file	VARCHAR(100) (Nullable)
qr_code_data	TEXT
email_sent	BOOLEAN
generated_at	DATETIME
regenerated_count	UNSIGNED INTEGER
booking_passenger_id	BIGINT (Foreign Key, Unique)

Bookings Booking

Field Name	Data Type
id	INTEGER (Primary Key)
pnr	VARCHAR(10) (Unique)
contact_name	VARCHAR(150)
contact_email	VARCHAR(254)
contact_phone	VARCHAR(20)

rebook_reason	VARCHAR(12)
travel_class	VARCHAR(60)
status	VARCHAR(20)
payment_amount	DECIMAL
payment_status	VARCHAR(10)
receipt_pdf	VARCHAR(100) (Nullable)
receipt_generated_at	DATETIME (Nullable)
booking_date	DATETIME
updated_at	DATETIME
flight_id	BIGINT (Foreign Key)
passenger_id	BIGINT (Foreign Key)
rebooked_from_id	BIGINT (Nullable, Self-referencing Foreign Key)

Bookings Booking passenger

Field Name	Data Type
id	INTEGER (Primary Key)
first_name	VARCHAR(100)
middle_name	VARCHAR(100)
last_name	VARCHAR(100)
date_of_birth	DATE (Nullable)
gender	VARCHAR(1)
nationality	VARCHAR(60)
email	VARCHAR(254)
phone_number	VARCHAR(20)
passenger_type	VARCHAR(10)
is_primary	BOOLEAN
seat_number	VARCHAR(6)
fare	DECIMAL
created_at	DATETIME
booking_id	BIGINT (Foreign Key)
seat_category_id	BIGINT (Nullable, Foreign Key)

Bookings Cancellation

Field Name	Data Type
id	INTEGER (Primary Key)
reason	TEXT
cancelled_at	DATETIME
time_before_departure_minutes	INTEGER
original_fare	DECIMAL

refund_eligible	BOOLEAN
pdf_file	VARCHAR(100) (Nullable)
booking_id	BIGINT (Foreign Key, Unique)

Bookings Refund

Field Name	Data Type
id	INTEGER (Primary Key)
amount	DECIMAL
status	VARCHAR(15)
created_at	DATETIME
processed_at	DATETIME (Nullable)
cancellation_id	BIGINT (Foreign Key, Unique)

Core Audit log

Field Name	Data Type
id	INTEGER (Primary Key)
module	VARCHAR(50)
action	VARCHAR(100)
description	TEXT
ip_address	CHAR(39) (Nullable)
timestamp	DATETIME
user_id	BIGINT (Nullable, Foreign Key)

Core System setting

Field Name	Data Type
id	INTEGER (Primary Key)
key	VARCHAR(100) (Unique)
value	VARCHAR(255)
description	VARCHAR(255)
updated_at	DATETIME

Flights Aircraft

Field Name	Data Type
id	INTEGER (Primary Key)
name	VARCHAR(100)

model	VARCHAR(100)
is_active	BOOLEAN
created_at	DATETIME

Flights Airport

Field Name	Data Type
id	INTEGER (Primary Key)
code	VARCHAR(3) (Unique)
name	VARCHAR(120)
city	VARCHAR(100)
state	VARCHAR(100)
country	VARCHAR(100)
address	VARCHAR(255)
contact_phone	VARCHAR(30)
contact_email	VARCHAR(254)
checkin_info	VARCHAR(255)
boarding_info	VARCHAR(255)
is_active	BOOLEAN

Flights Delay record

Field Name	Data Type
id	INTEGER (Primary Key)
previous_delay_minutes	UNSIGNED INTEGER
additional_delay_minutes	UNSIGNED INTEGER
total_delay_minutes	UNSIGNED INTEGER
facility_level	VARCHAR(25)
reason	VARCHAR(255)
created_at	DATETIME
updated_by_id	BIGINT (Nullable, Foreign Key)
flight_id	BIGINT (Foreign Key)

Flights Flight

Field Name	Data Type
id	INTEGER (Primary Key)
flight_number	VARCHAR(10) (Unique)
airline_name	VARCHAR(100)
departure_date	DATE
departure_time	TIME

arrival_date	DATE
arrival_time	TIME
original_departure_datetime	DATETIME (Nullable)
updated_departure_datetime	DATETIME (Nullable)
delay_minutes	UNSIGNED INTEGER
delay_reason	VARCHAR(255)
cancellation_reason	VARCHAR(255)
origin_gate_id	BIGINT (Nullable, Foreign Key)
destination_terminal_id	BIGINT (Nullable, Foreign Key)
destination_gate_id	BIGINT (Nullable, Foreign Key)
boarding_time	TIME (Nullable)
boarding_passes_generated_at	DATETIME (Nullable)
seat_capacity	UNSIGNED INTEGER
available_seats	UNSIGNED INTEGER
status	VARCHAR(20)
created_at	DATETIME
updated_at	DATETIME
aircraft_id	BIGINT (Nullable, Foreign Key)
destination_id	BIGINT (Foreign Key)
origin_id	BIGINT (Foreign Key)
origin_terminal_id	BIGINT (Nullable, Foreign Key)

Flights Gate

Field Name	Data Type
id	INTEGER (Primary Key)
name	VARCHAR(60)
created_at	DATETIME
terminal_id	BIGINT (Foreign Key)

Flights Seat

Field Name	Data Type
id	INTEGER (Primary Key)
seat_number	VARCHAR(6)
row_number	UNSIGNED INTEGER
column_letter	VARCHAR(1)
is_blocked	BOOLEAN
aircraft_id	BIGINT (Foreign Key)
category_id	BIGINT (Foreign Key)

Flights Seat category

Field Name	Data Type
id	INTEGER (Primary Key)
name	VARCHAR(60)
code	VARCHAR(30)
display_order	UNSIGNED INTEGER
price_multiplier	DECIMAL
color	VARCHAR(20)
aircraft_id	BIGINT (Foreign Key)

Flights Terminal

Field Name	Data Type
id	INTEGER (Primary Key)
name	VARCHAR(60)
created_at	DATETIME
airport_id	BIGINT (Foreign Key)

Notifications Notification

Field Name	Data Type
id	INTEGER (Primary Key)
notification_type	VARCHAR(20)
title	VARCHAR(150)
message	TEXT
is_read	BOOLEAN
created_at	DATETIME
recipient_id	BIGINT (Foreign Key)
related_booking_id	BIGINT (Nullable, Foreign Key)

Recovery Disruption record

Field Name	Data Type
id	INTEGER (Primary Key)
disruption_type	VARCHAR(15)
delay_minutes	UNSIGNED INTEGER
reason	VARCHAR(255)
eligible_for_recovery	BOOLEAN
eligible_for_rebooking	BOOLEAN
eligible_for_refund	BOOLEAN

eligible_for_meal_voucher	BOOLEAN
eligible_for_hotel_voucher	BOOLEAN
eligible_for_transport_voucher	BOOLEAN
resolved	BOOLEAN
created_at	DATETIME
booking_id	BIGINT (Foreign Key)
flight_id	BIGINT (Foreign Key)

Recovery Passenger facility

Field Name	Data Type
id	INTEGER (Primary Key)
facility_type	VARCHAR(12)
status	VARCHAR(10)
created_at	DATETIME
updated_at	DATETIME
booking_id	BIGINT (Foreign Key)
voucher_id	BIGINT (Nullable, Unique, Foreign Key)

Recovery Rebooking request

Field Name	Data Type
id	INTEGER (Primary Key)
status	VARCHAR(12)
staff_remarks	TEXT
requested_at	DATETIME
processed_at	DATETIME (Nullable)
new_booking_id	BIGINT (Nullable, Unique, Foreign Key)
original_booking_id	BIGINT (Foreign Key)
passenger_id	BIGINT (Foreign Key)
processed_by_id	BIGINT (Nullable, Foreign Key)
requested_flight_id	BIGINT (Foreign Key)

Recovery Refund request

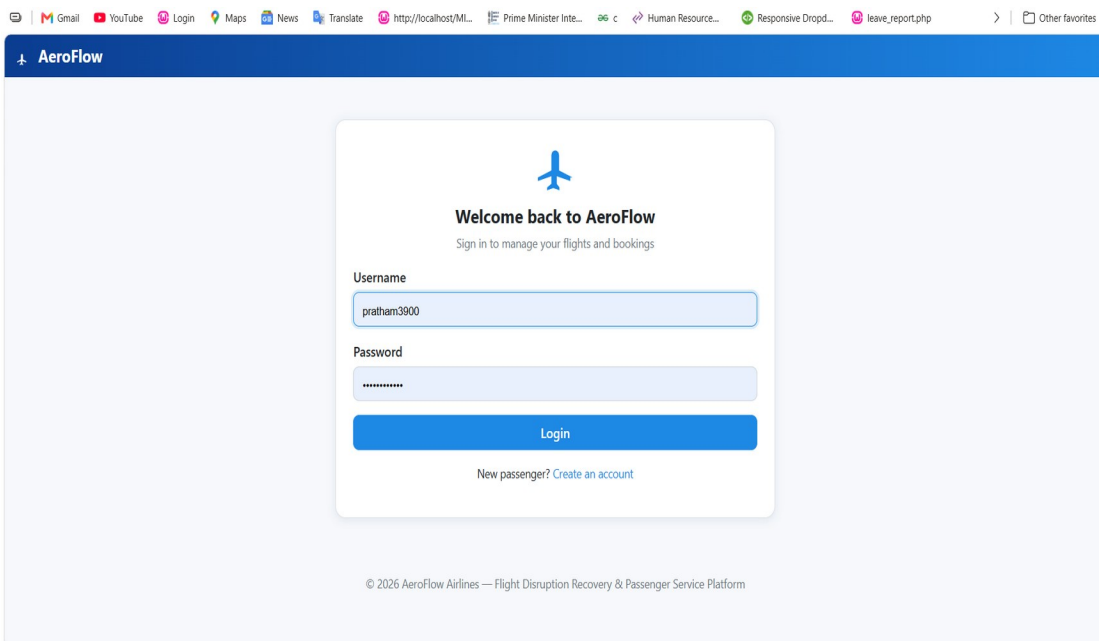
Field Name	Data Type
id	INTEGER (Primary Key)
refund_id	VARCHAR(16) (Unique)
reason	TEXT
requested_amoun	DECIMAL

t	
status	VARCHAR(12)
staff_remarks	TEXT
request_date	DATETIME
processed_date	DATETIME (Nullable)
booking_id	BIGINT (Foreign Key)
passenger_id	BIGINT (Foreign Key)
processed_by_id	BIGINT (Nullable, Foreign Key)

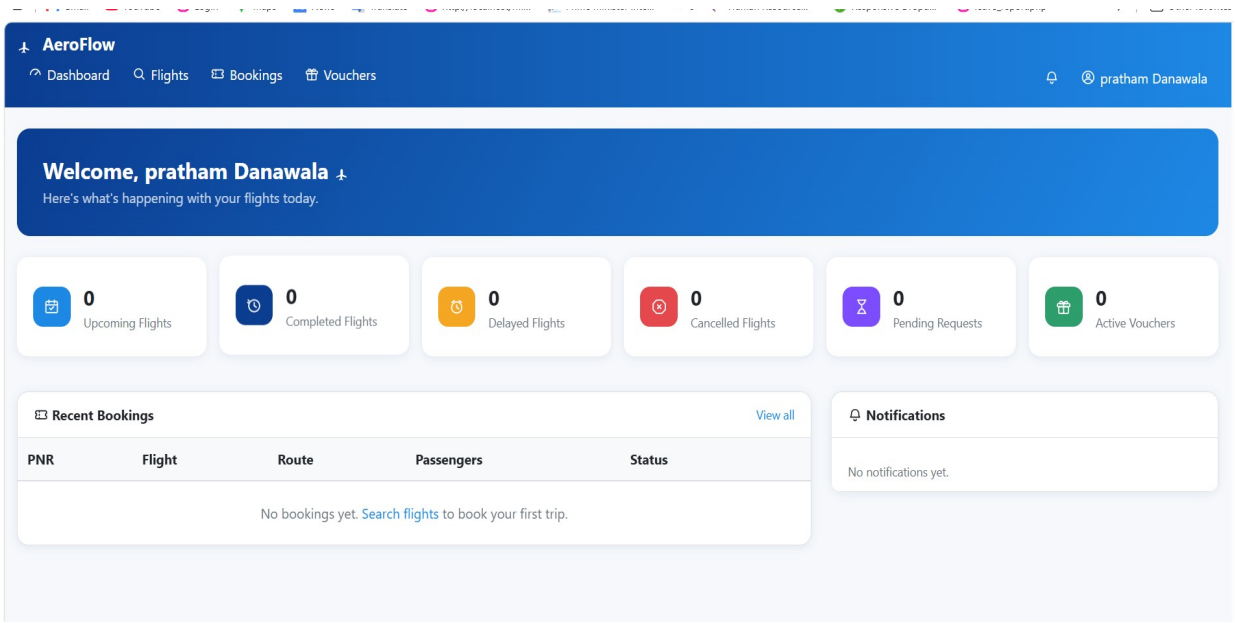
Vouchers Voucher

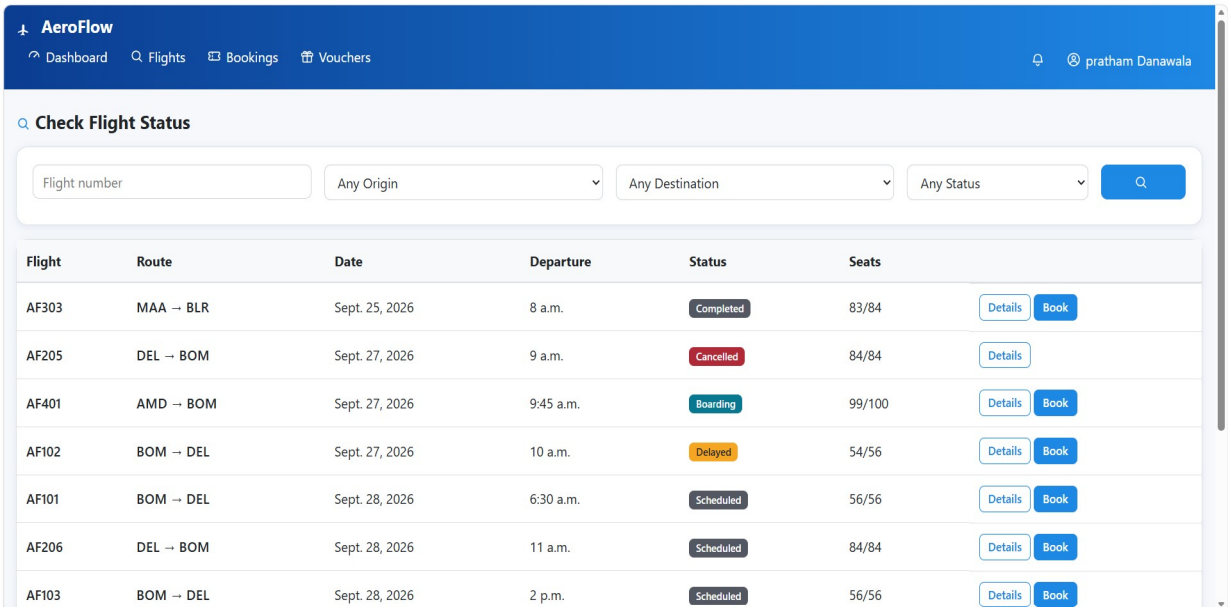
Field Name	Data Type
id	INTEGER (Primary Key)
voucher_code	VARCHAR(16) (Unique)
voucher_type	VARCHAR(12)
amount	DECIMAL
issue_date	DATETIME
expiry_date	DATE
status	VARCHAR(12)
usage_instructions	TEXT
pdf_file	VARCHAR(100) (Nullable)
booking_id	BIGINT (Foreign Key)
passenger_id	BIGINT (Foreign Key)

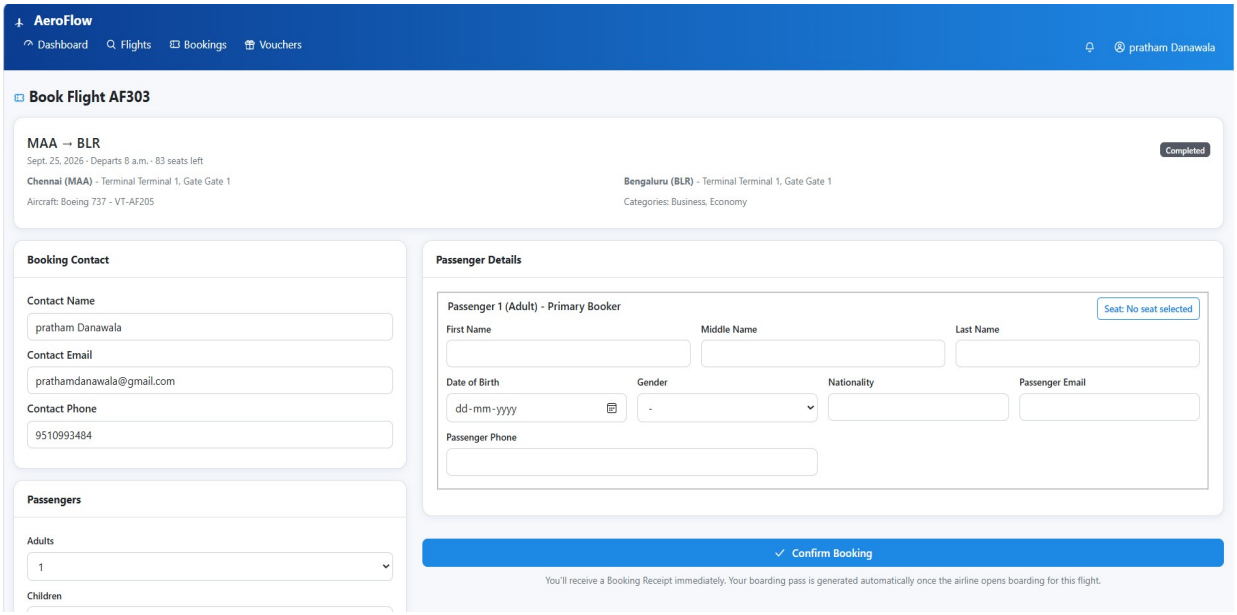
➢ Screen Layouts

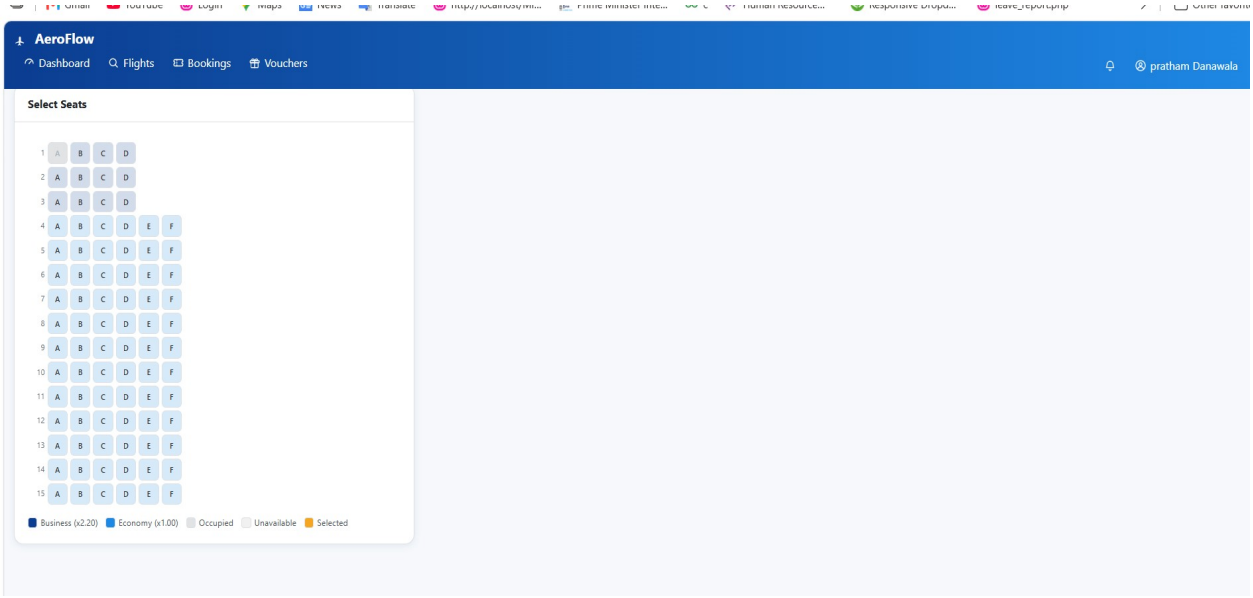


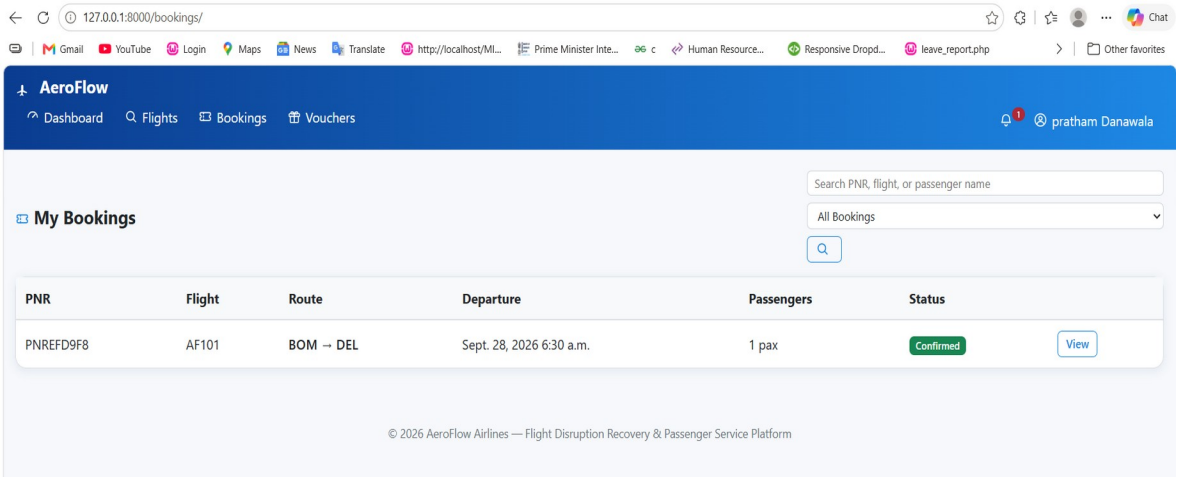
Passenger Dashboard

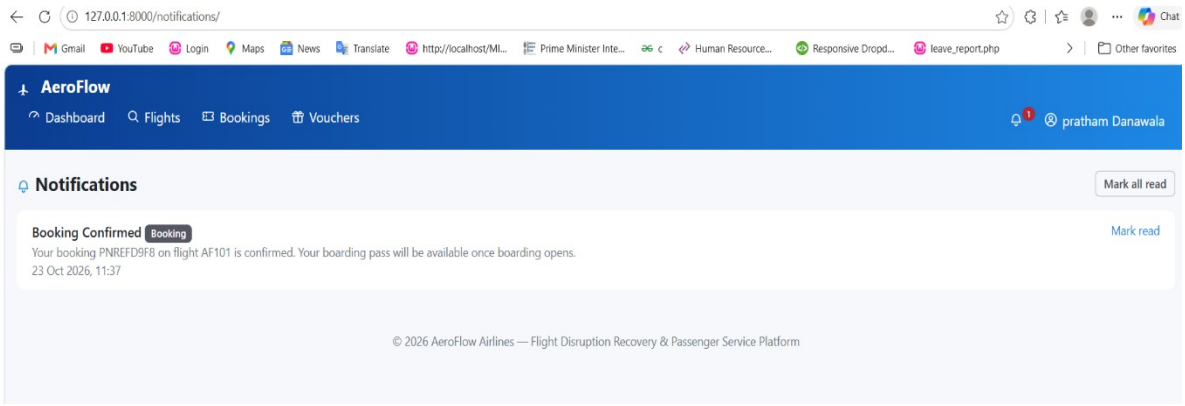


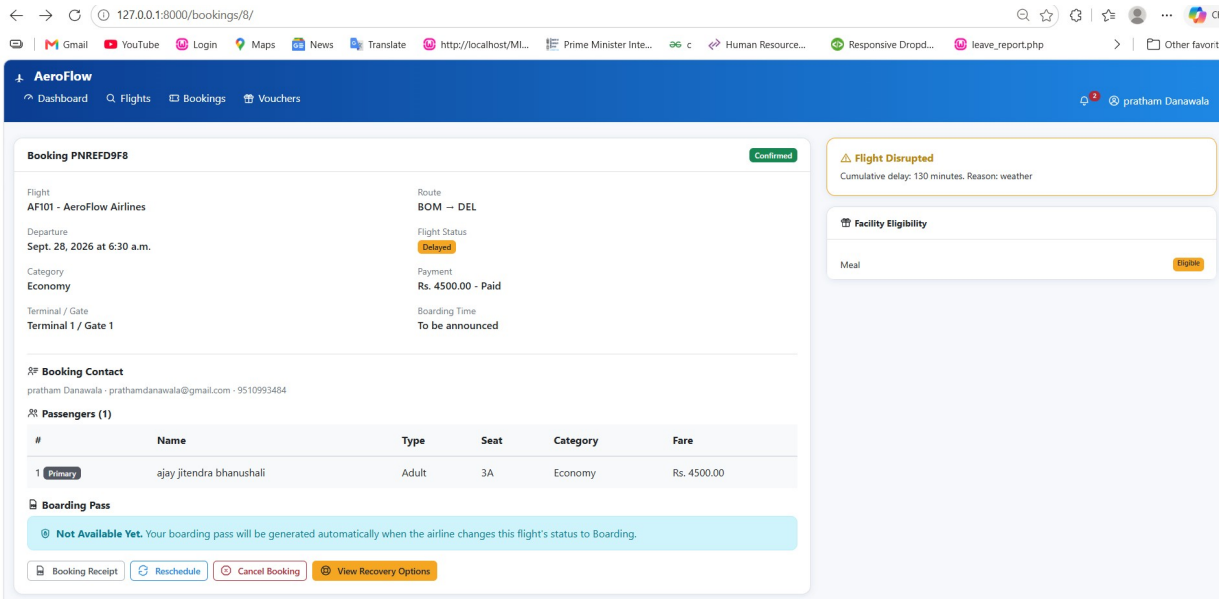

AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
Check Flight Status
Flight number
Any Origin
Any Destination
Any Status
Flight
Route
Date
Departure
Status
Seats
AF303
MAA → BLR
Sept. 25, 2026
8 a.m.
Completed
83/84
Details
Book
AF205
DEL → BOM
Sept. 27, 2026
9 a.m.
Cancelled
84/84
Details
AF401
AMD → BOM
Sept. 27, 2026
9:45 a.m.
Boarding
99/100
Details
Book
AF102
BOM → DEL
Sept. 27, 2026
10 a.m.
Delayed
54/56
Details
Book
AF101
BOM → DEL
Sept. 28, 2026
6:30 a.m.
Scheduled
56/56
Details
Book
AF206
DEL → BOM
Sept. 28, 2026
11 a.m.
Scheduled
84/84
Details
Book
AF103
BOM → DEL
Sept. 28, 2026
2 p.m.
Scheduled
56/56
Details
Book

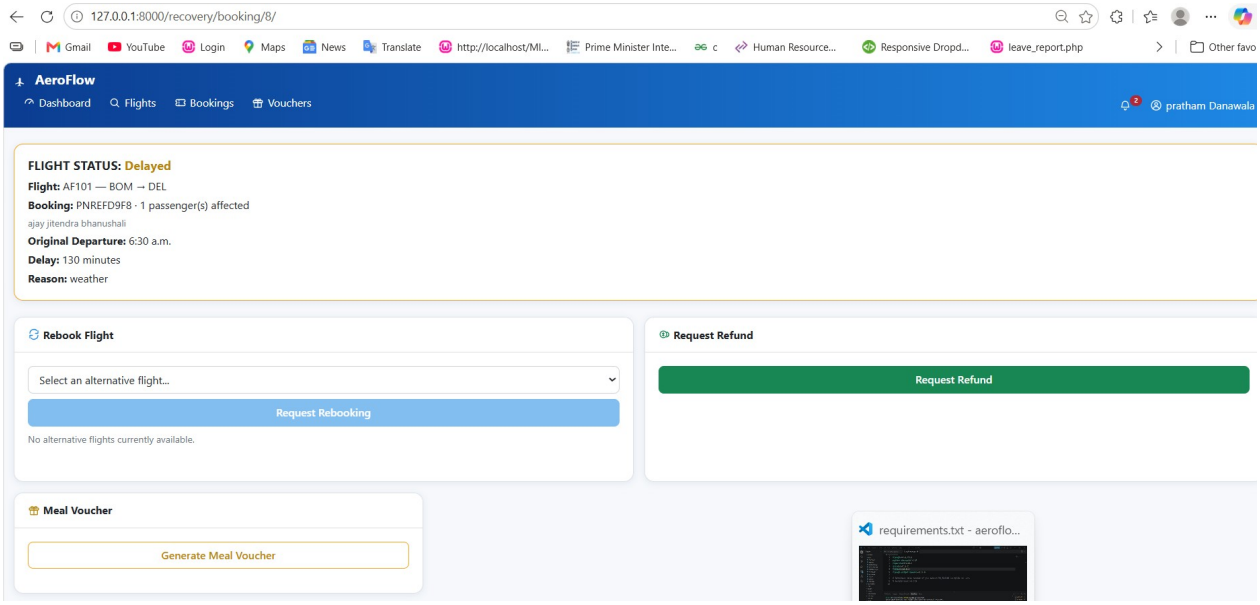

AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
Book Flight AF303
MAA → BLR
Sept. 25, 2026 · Departs 8 a.m. · 83 seats left
Completed
Chennai (MAA) - Terminal Terminal 1, Gate Gate 1
Aircraft: Boeing 737 - VT-AF205
Bengaluru (BLR) - Terminal Terminal 1, Gate Gate 1
Categories: Business, Economy
Booking Contact
Contact Name
pratham Danawala
Contact Email
prathamdanawala@gmail.com
Contact Phone
9510993484
Passengers
Adults
1
Children
Passenger Details
Passenger 1 (Adult) - Primary Booker
Seat: No seat selected
First Name
Middle Name
Last Name
Date of Birth
dd-mm-yyyy
Gender
Nationality
Passenger Email
Passenger Phone
Confirm Booking
You'll receive a Booking Receipt immediately. Your boarding pass is generated automatically once the airline opens boarding for this flight.


AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
Select Seats
Business (x2.20)
Economy (x1.00)
Occupied
Unavailable
Selected

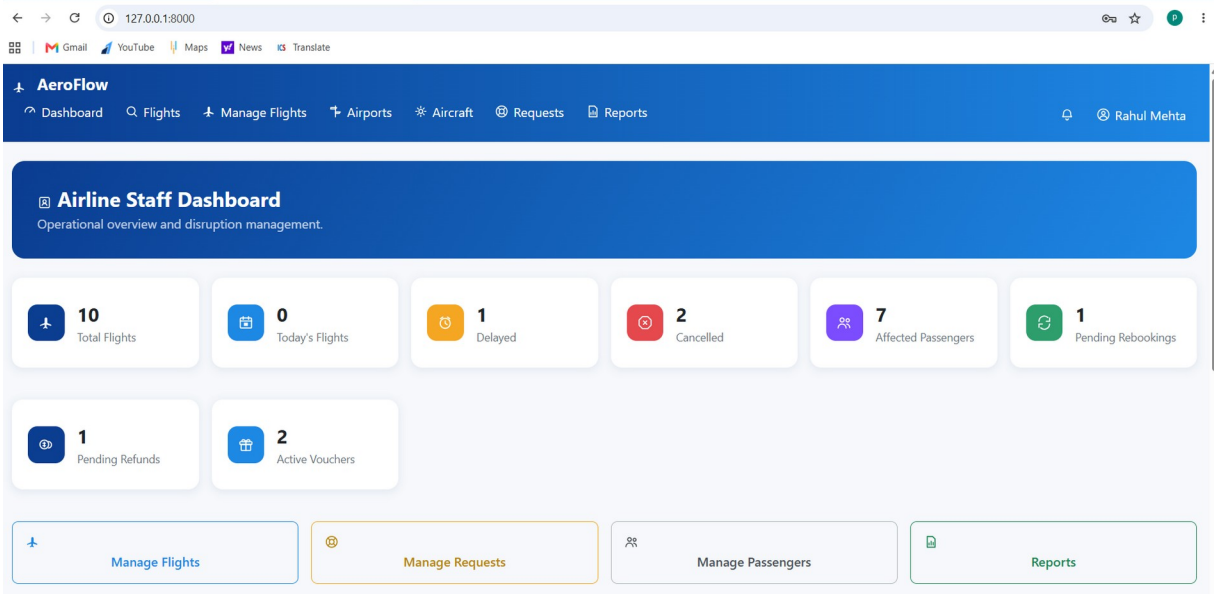

127.0.0.1:8000/bookings/
AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
My Bookings
Search PNR, flight, or passenger name
All Bookings
PNR
Flight
Route
Departure
Passengers
Status
PNREFD9F8
AF101
BOM → DEL
Sept. 28, 2026 6:30 a.m.
1 pax
Confirmed
View
© 2026 AeroFlow Airlines — Flight Disruption Recovery & Passenger Service Platform

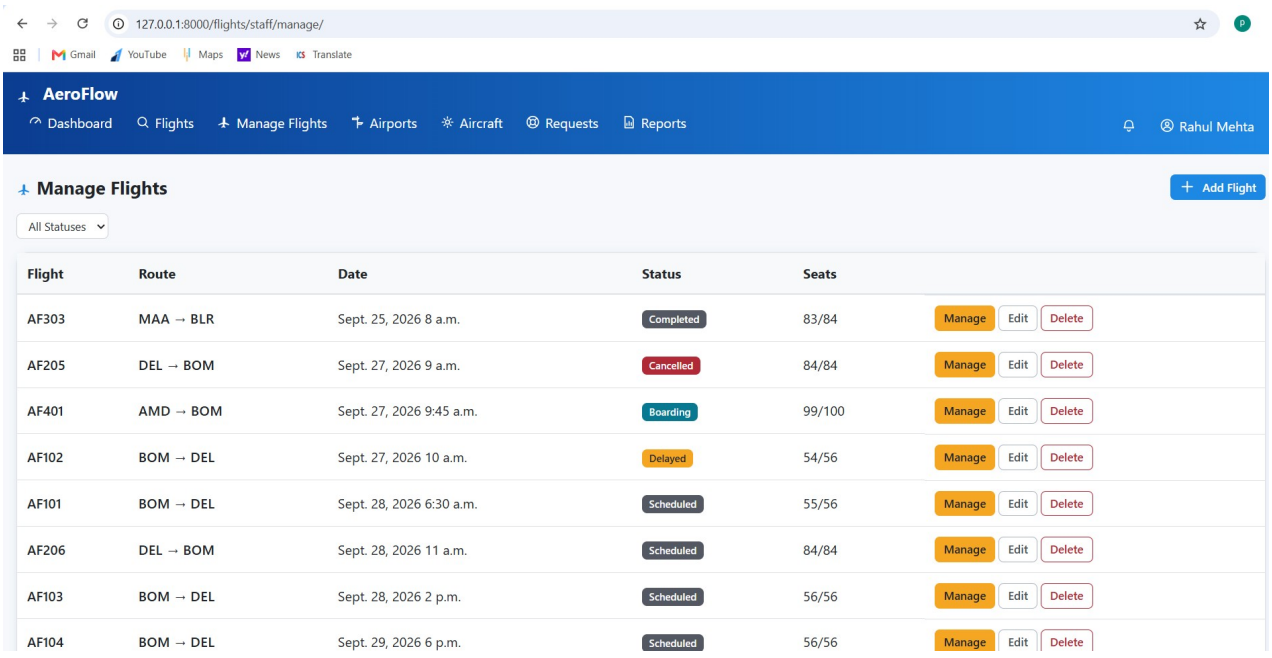

127.0.0.1:8000/notifications/
AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
Notifications
Mark all read
Booking Confirmed
Booking
Mark read
Your booking PNREFD9F8 on flight AF101 is confirmed. Your boarding pass will be available once boarding opens.
23 Oct 2026, 11:37
© 2026 AeroFlow Airlines — Flight Disruption Recovery & Passenger Service Platform

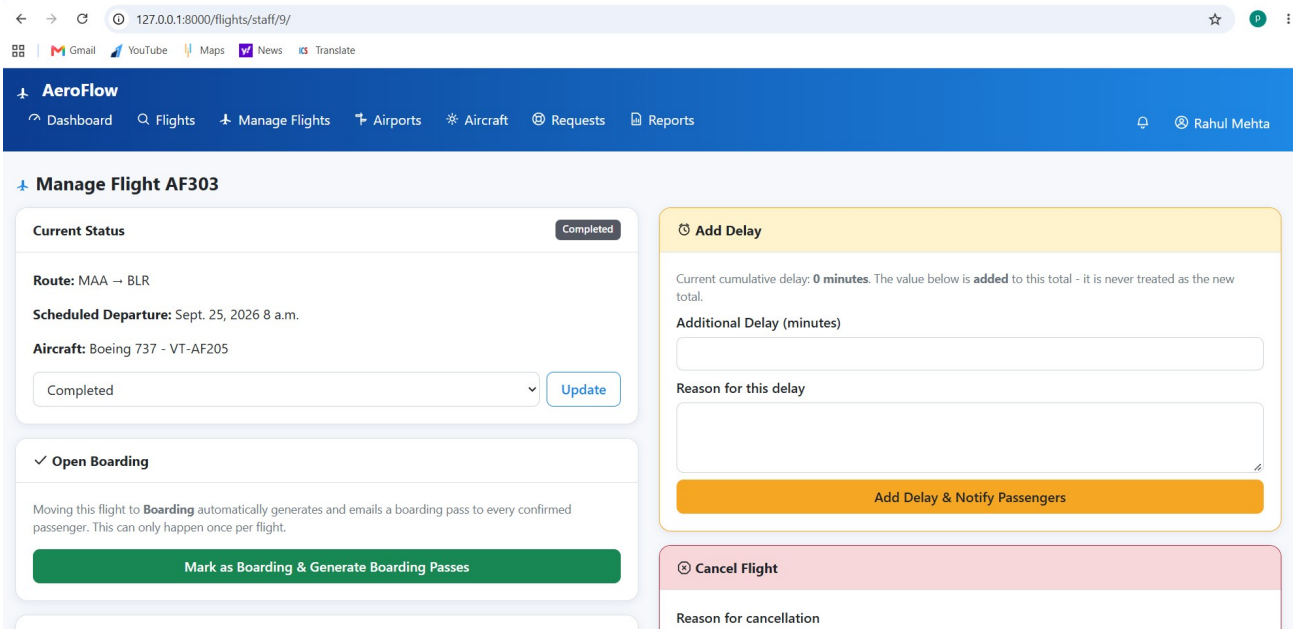

127.0.0.1:8000/bookings/8/
AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
Booking PNREFD9F8
Confirmed
Flight
AF101 - AeroFlow Airlines
Route
BOM → DEL
Departure
Sept. 28, 2026 at 6:30 a.m.
Flight Status
Delayed
Category
Economy
Payment
Rs. 4500.00 - Paid
Terminal / Gate
Terminal 1 / Gate 1
Boarding Time
To be announced
Booking Contact
pratham Danawala · prathamdanawala@gmail.com · 9510993484
Passengers (1)
Name Type Seat Category Fare
1 Primary ajay jitendra bhanushali Adult 3A Economy Rs. 4500.00
Boarding Pass
Not Available Yet. Your boarding pass will be generated automatically when the airline changes this flight's status to Boarding.
Booking Receipt
Reschedule
Cancel Booking
View Recovery Options
Flight Disrupted
Cumulative delay: 130 minutes. Reason: weather
Facility Eligibility
Meal
Eligible

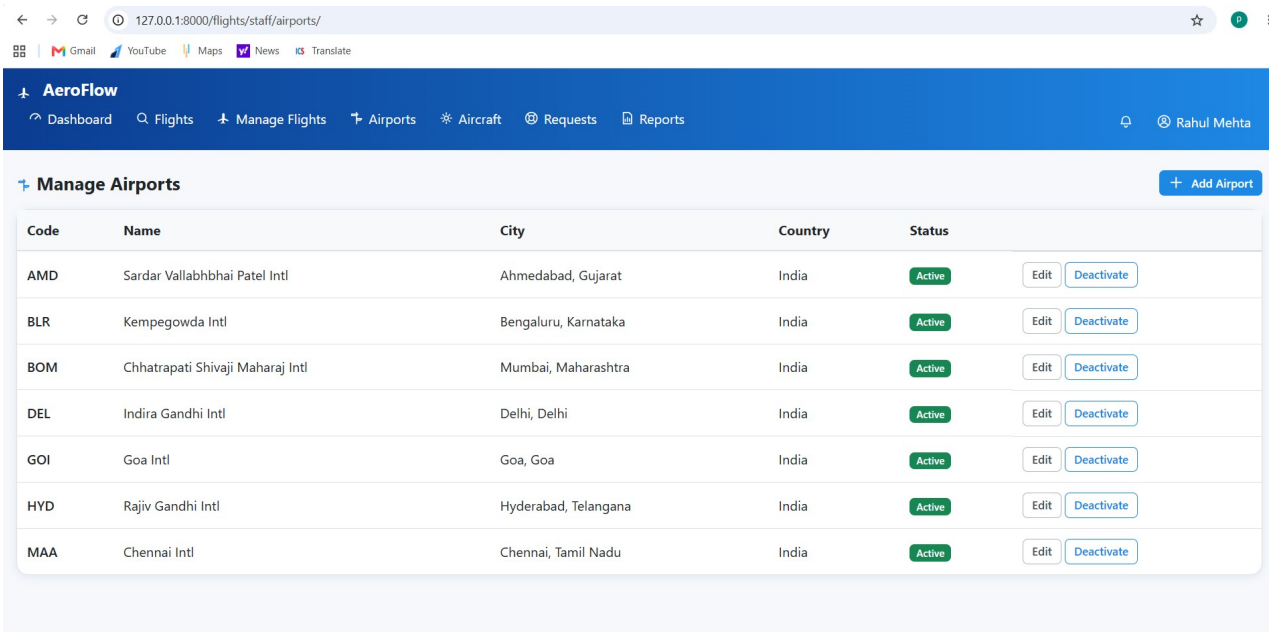

127.0.0.1:8000/recovery/booking/8/
AeroFlow
Dashboard
Flights
Bookings
Vouchers
pratham Danawala
FLIGHT STATUS: Delayed
Flight: AF101 — BOM → DEL
Booking: PNREFD9F8 · 1 passenger(s) affected
ajay jitendra bhanushali
Original Departure: 6:30 a.m.
Delay: 130 minutes
Reason: weather
Rebook Flight
Select an alternative flight...
Request Rebooking
No alternative flights currently available.
Request Refund
Request Refund
Meal Voucher
Generate Meal Voucher
requirements.txt - aeroflo...

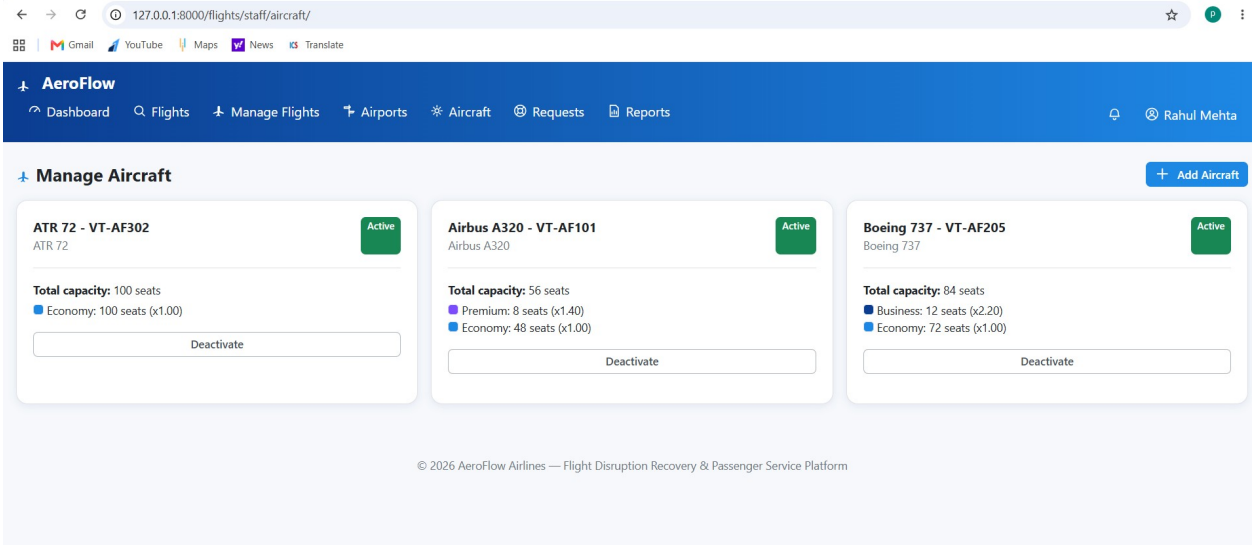
Staff Dashboard

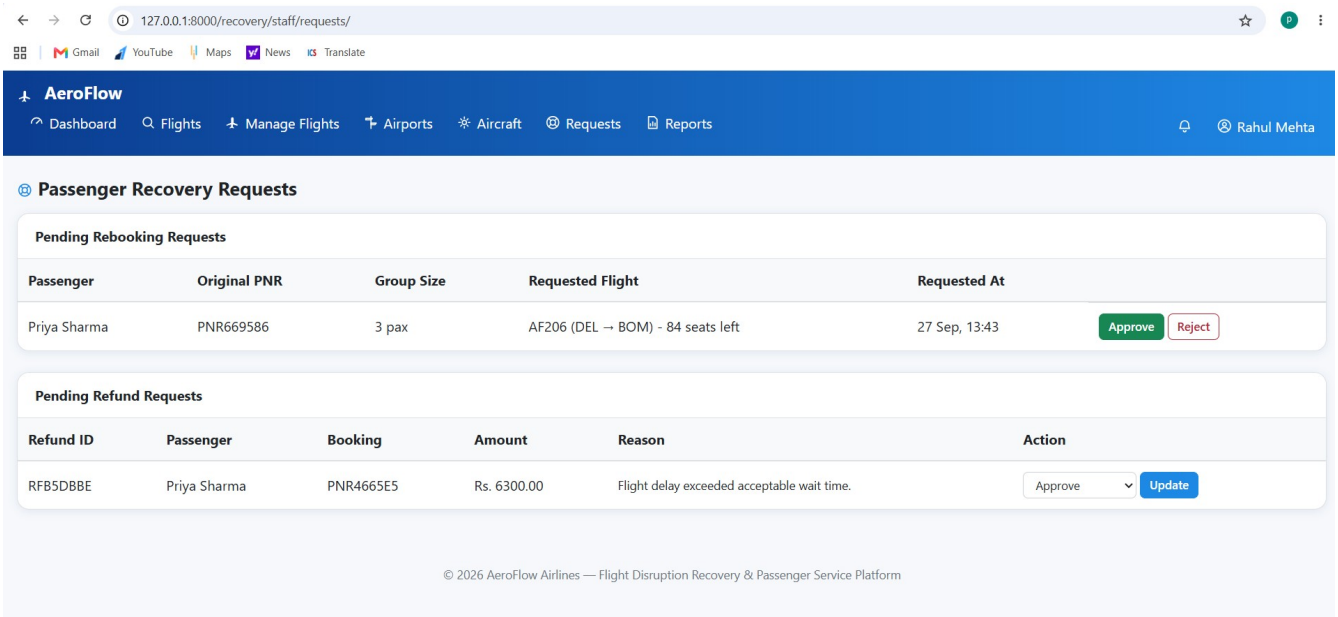


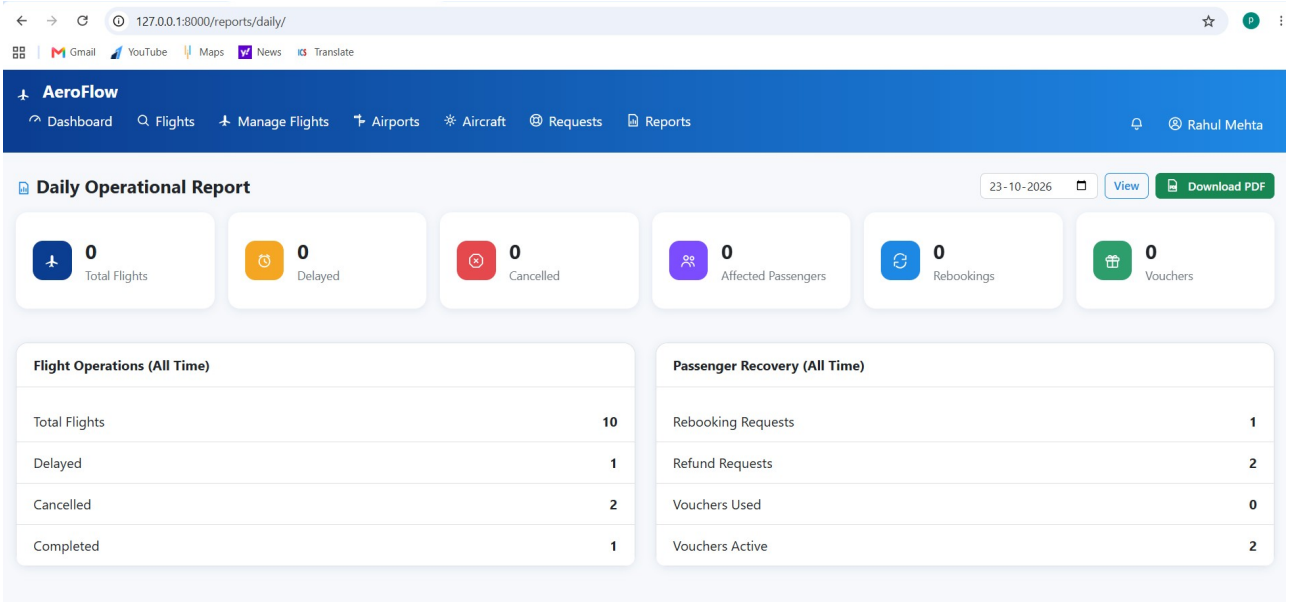



127.0.0.1:8000/flights/staff/9/
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Rahul Mehta
Manage Flight AF303
Current Status
Completed
Route: MAA → BLR
Scheduled Departure: Sept. 25, 2026 8 a.m.
Aircraft: Boeing 737 - VT-AF205
Completed
Update
Open Boarding
Moving this flight to Boarding automatically generates and emails a boarding pass to every confirmed passenger. This can only happen once per flight.
Mark as Boarding & Generate Boarding Passes
Add Delay
Current cumulative delay: 0 minutes. The value below is added to this total - it is never treated as the new total.
Additional Delay (minutes)
Reason for this delay
Add Delay & Notify Passengers
Cancel Flight
Reason for cancellation

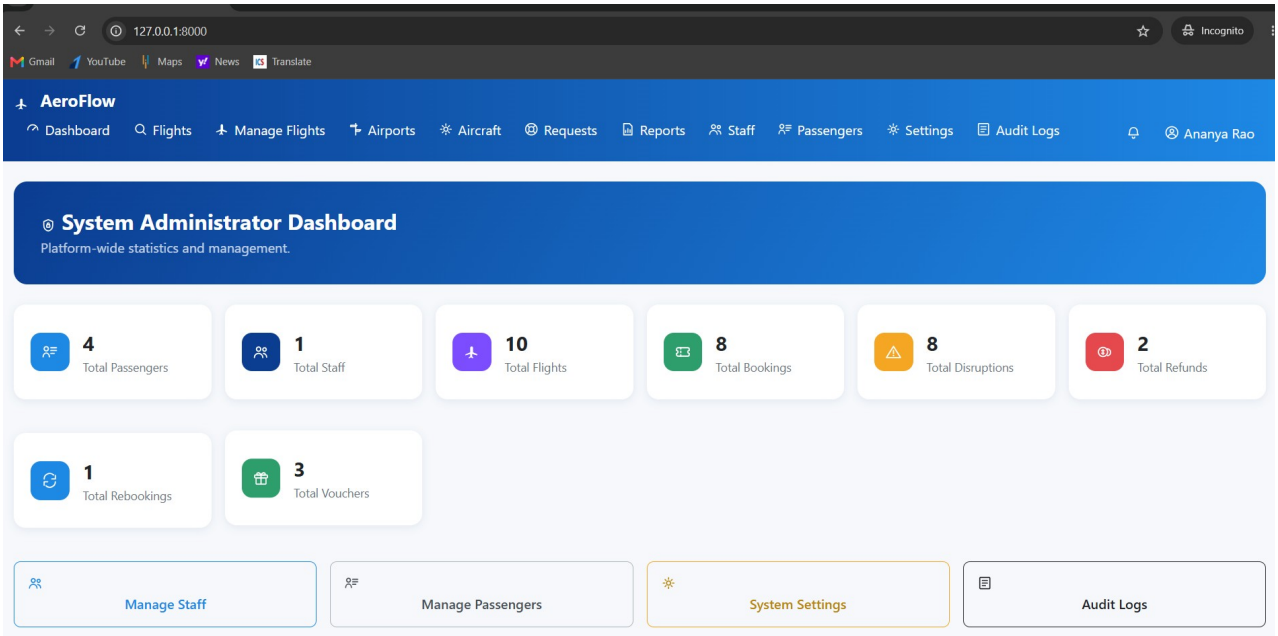

127.0.0.1:8000/flights/staff/airports/
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Rahul Mehta
Manage Airports
Add Airport
Code
Name
City
Country
Status
AMD
Sardar Vallabhbhai Patel Intl
Ahmedabad, Gujarat
India
Active
Edit
Deactivate
BLR
Kempegowda Intl
Bengaluru, Karnataka
India
Active
Edit
Deactivate
BOM
Chhatrapati Shivaji Maharaj Intl
Mumbai, Maharashtra
India
Active
Edit
Deactivate
DEL
Indira Gandhi Intl
Delhi, Delhi
India
Active
Edit
Deactivate
GOI
Goa Intl
Goa, Goa
India
Active
Edit
Deactivate
HYD
Rajiv Gandhi Intl
Hyderabad, Telangana
India
Active
Edit
Deactivate
MAA
Chennai Intl
Chennai, Tamil Nadu
India
Active
Edit
Deactivate

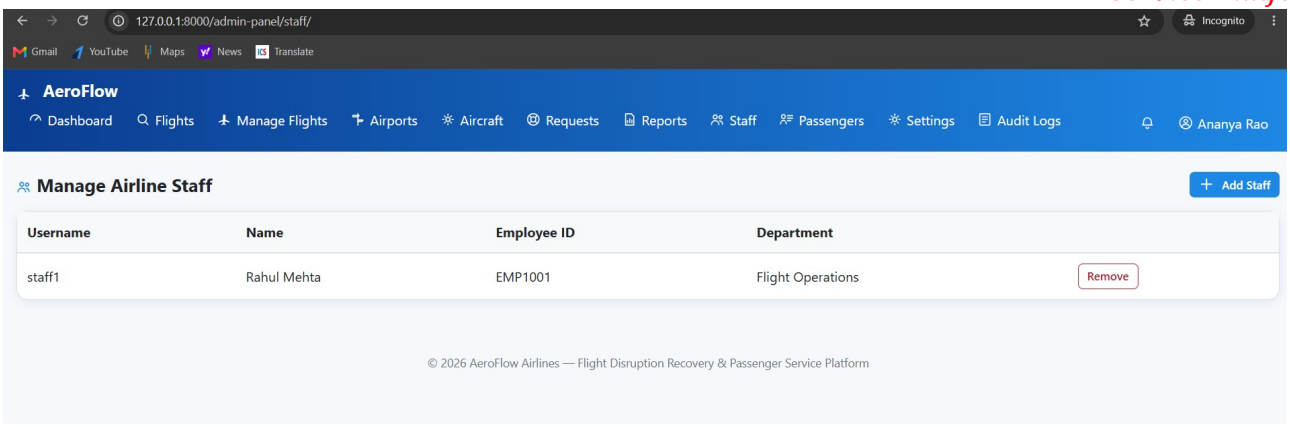

127.0.0.1:8000/flights/staff/aircraft/
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Rahul Mehta
Manage Aircraft
Add Aircraft
ATR 72 - VT-AF302
ATR 72
Active
Total capacity: 100 seats
Economy: 100 seats (x1.00)
Deactivate
Airbus A320 - VT-AF101
Airbus A320
Active
Total capacity: 56 seats
Premium: 8 seats (x1.40)
Economy: 48 seats (x1.00)
Deactivate
Boeing 737 - VT-AF205
Boeing 737
Active
Total capacity: 84 seats
Business: 12 seats (x2.20)
Economy: 72 seats (x1.00)
Deactivate
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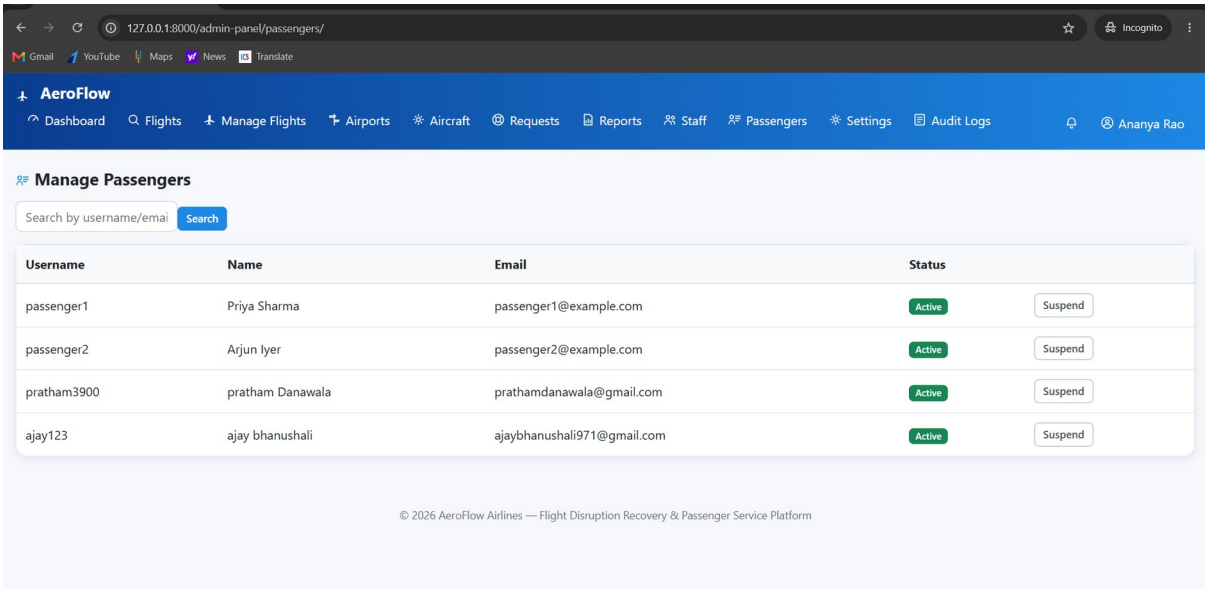

127.0.0.1:8000/recovery/staff/requests/
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Rahul Mehta
Passenger Recovery Requests
Pending Rebooking Requests
Passenger
Original PNR
Group Size
Requested Flight
Requested At
Priya Sharma
PNR669586
3 pax
AF206 (DEL → BOM) - 84 seats left
27 Sep, 13:43
Approve
Reject
Pending Refund Requests
Refund ID
Passenger
Booking
Amount
Reason
Action
RFB5DBBE
Priya Sharma
PNR4665E5
Rs. 6300.00
Flight delay exceeded acceptable wait time.
Approve
Update
© 2026 AeroFlow Airlines — Flight Disruption Recovery & Passenger Service Platform

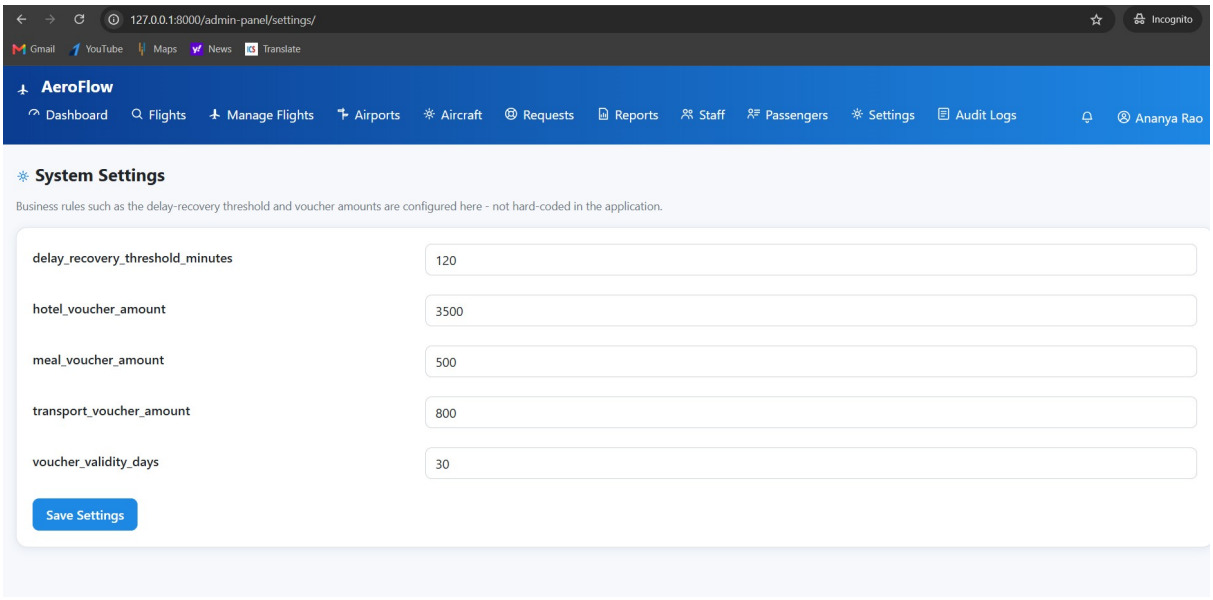


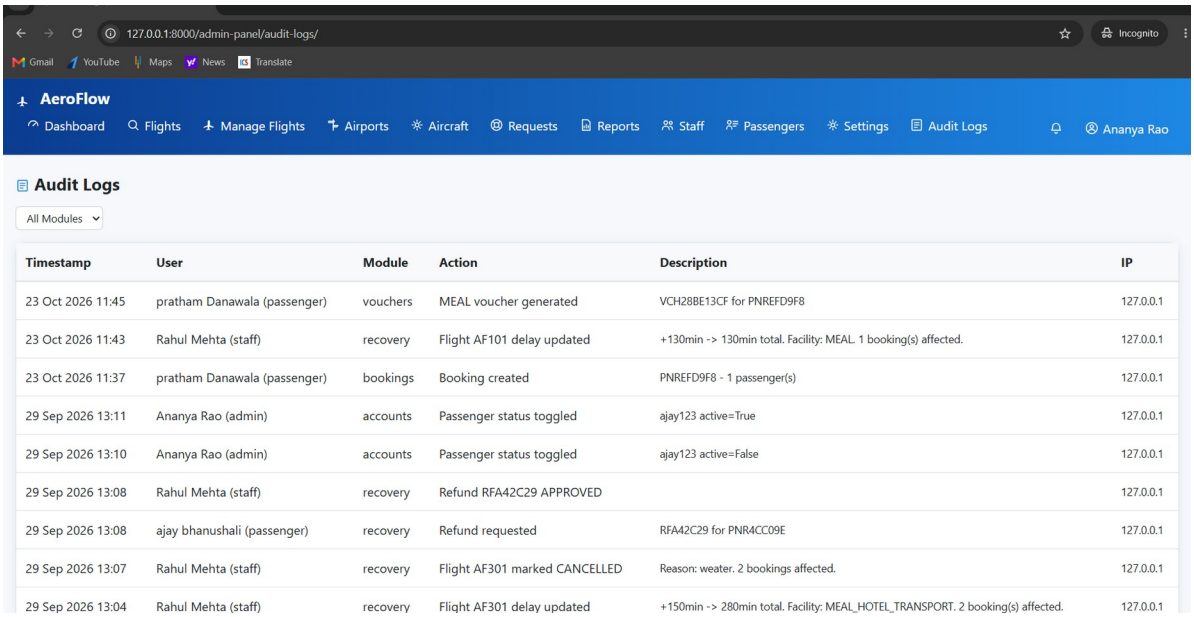
Admin Dashboard




127.0.0.1:8000/admin-panel/staff/
Incognito
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Staff
Passengers
Settings
Audit Logs
Ananya Rao
Manage Airline Staff
Add Staff
Username
Name
Employee ID
Department
staff1
Rahul Mehta
EMP1001
Flight Operations
Remove
© 2026 AeroFlow Airlines — Flight Disruption Recovery & Passenger Service Platform

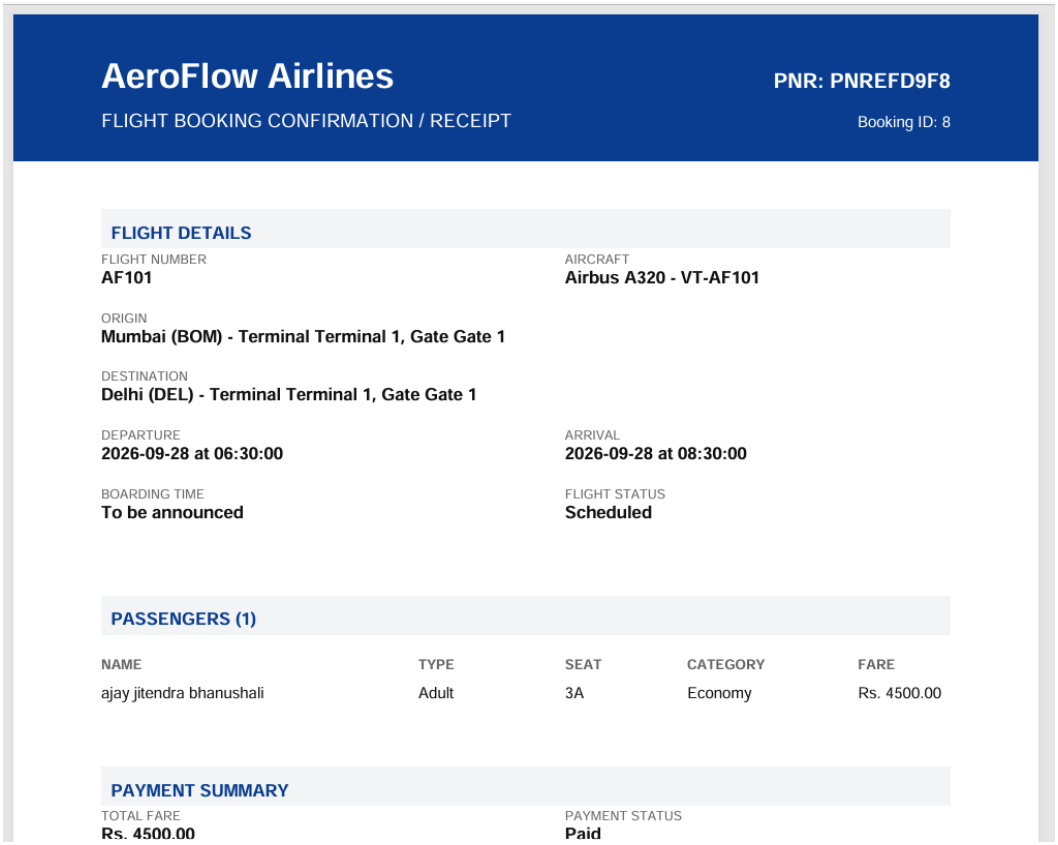

127.0.0.1:8000/admin-panel/passengers/
Incognito
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Staff
Passengers
Settings
Audit Logs
Ananya Rao
Manage Passengers
Search by username/emai
Search
Username
Name
Email
Status
passenger1
Priya Sharma
passenger1@example.com
Active
Suspend
passenger2
Arjun Iyer
passenger2@example.com
Active
Suspend
pratham3900
pratham Danawala
prathamdanawala@gmail.com
Active
Suspend
ajay123
ajay bhanushali
ajaybhanushali971@gmail.com
Active
Suspend
© 2026 AeroFlow Airlines — Flight Disruption Recovery & Passenger Service Platform


127.0.0.1:8000/admin-panel/settings/
Incognito
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Staff
Passengers
Settings
Audit Logs
Ananya Rao
System Settings
Business rules such as the delay-recovery threshold and voucher amounts are configured here - not hard-coded in the application.
delay_recovery_threshold_minutes
120
hotel_voucher_amount
3500
meal_voucher_amount
500
transport_voucher_amount
800
voucher_validity_days
30
Save Settings


127.0.0.1:8000/admin-panel/audit-logs/
Incognito
Gmail
YouTube
Maps
News
Translate
AeroFlow
Dashboard
Flights
Manage Flights
Airports
Aircraft
Requests
Reports
Staff
Passengers
Settings
Audit Logs
Ananya Rao
Audit Logs
All Modules
Timestamp
User
Module
Action
Description
IP
23 Oct 2026 11:45
pratham Danawala (passenger)
vouchers
MEAL voucher generated
VCH28BE13CF for PNREFD9F8
127.0.0.1
23 Oct 2026 11:43
Rahul Mehta (staff)
recovery
Flight AF101 delay updated
+130min -> 130min total. Facility: MEAL. 1 booking(s) affected.
127.0.0.1
23 Oct 2026 11:37
pratham Danawala (passenger)
bookings
Booking created
PNREFD9F8 - 1 passenger(s)
127.0.0.1
29 Sep 2026 13:11
Ananya Rao (admin)
accounts
Passenger status toggled
ajay123 active=True
127.0.0.1
29 Sep 2026 13:10
Ananya Rao (admin)
accounts
Passenger status toggled
ajay123 active=False
127.0.0.1
29 Sep 2026 13:08
Rahul Mehta (staff)
recovery
Refund RFA42C29 APPROVED
127.0.0.1
29 Sep 2026 13:08
ajay bhanushali (passenger)
recovery
Refund requested
RFA42C29 for PNR4CC09E
127.0.0.1
29 Sep 2026 13:07
Rahul Mehta (staff)
recovery
Flight AF301 marked CANCELLED
Reason: weater. 2 bookings affected.
127.0.0.1
29 Sep 2026 13:04
Rahul Mehta (staff)
recovery
Flight AF301 delay updated
+150min -> 280min total. Facility: MEAL_HOTEL_TRANSPORT. 2 booking(s) affected.
127.0.0.1

➢ Reports

Booking Receipt



AeroFlow Airlines

FLIGHT BOOKING CONFIRMATION / RECEIPT

PNR: PNREFD9F8

Booking ID: 8

FLIGHT DETAILS

FLIGHT NUMBER
AF101

AIRCRAFT
Airbus A320 - VT-AF101

ORIGIN
Mumbai (BOM) - Terminal Terminal 1, Gate Gate 1

DESTINATION
Delhi (DEL) - Terminal Terminal 1, Gate Gate 1

DEPARTURE
2026-09-28 at 06:30:00

ARRIVAL
2026-09-28 at 08:30:00

BOARDING TIME
To be announced

FLIGHT STATUS
Scheduled

PASSENGERS (1)

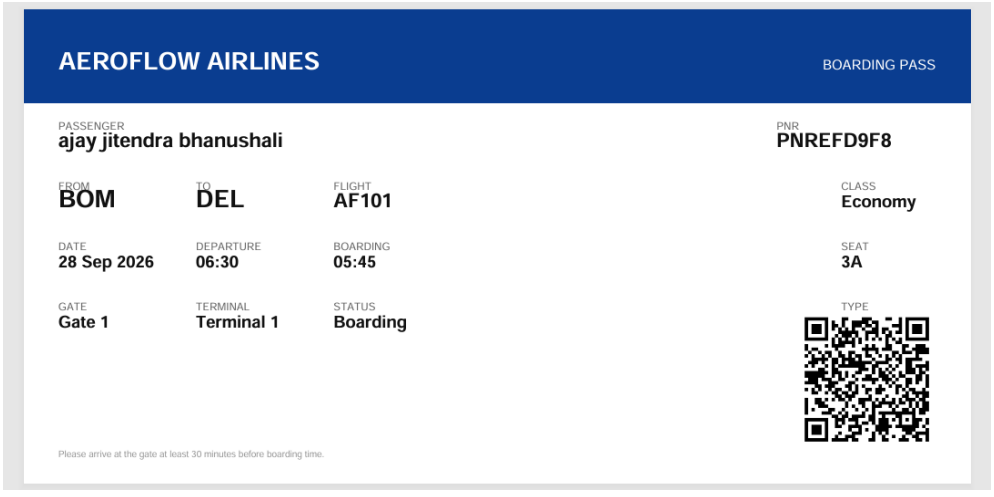
NAME	TYPE	SEAT	CATEGORY	FARE
ajay jitendra bhanushali	Adult	3A	Economy	Rs. 4500.00

PAYMENT SUMMARY

TOTAL FARE
Rs. 4500.00

PAYMENT STATUS
Paid

Boarding pass



AEROFLOW AIRLINES

BOARDING PASS

PASSENGER
ajay jitendra bhanushali

PNR
PNREFD9F8

FROM
BOM

TO
DEL

FLIGHT
AF101

CLASS
Economy

DATE
28 Sep 2026

DEPARTURE
06:30

BOARDING
05:45

SEAT
3A

GATE
Gate 1

TERMINAL
Terminal 1

STATUS
Boarding

TYPE

Please arrive at the gate at least 30 minutes before boarding time.

Voucher

AeroFlow Airlines

MEAL VOUCHER

Voucher Code: VCH28BE13CF

Passenger: pratham Danawala

Booking PNR: PNREFD9F8

Value: Rs. 500.0

Issued: 23 Oct 2026 Expiry: 22 Nov 2026

Status: Active

Present this voucher at any airport restaurant partner.

Operational Report

AeroFlow - Daily Operational Report

Date: 23 October 2026

Metric	Value
Total Flights	0
Delayed Flights	0
Cancelled Flights	0
Completed Flights	0
Affected Passengers	1
Rebookings	0
Refunds	0
Vouchers Issued	1

System Testing

6.1 Introduction

System testing is an important phase of software development in which the application is tested to verify that its features work according to the specified requirements. It helps identify errors, validate system functionality, and ensure that the application behaves as expected.

AeroFlow is tested to verify its passenger services, flight management operations, booking functionality, and disruption recovery workflows. Testing also helps ensure that the application handles invalid inputs and unexpected situations appropriately.

6.2 Testing Strategies

The following testing strategies can be used to test the AeroFlow application.

6.2.1 Unit Testing

Unit testing checks individual components or functions of an application separately. In AeroFlow, individual functions related to booking operations, flight management, validation, and other business logic can be tested independently.

6.2.2 Integration Testing

Integration testing verifies whether different modules work correctly together. In AeroFlow, it can be used to test interactions between passenger accounts, flight records, bookings, email notifications, and document generation.

6.2.3 Functional Testing

Functional testing verifies whether the application's features work according to their requirements. In AeroFlow, this includes testing login, flight search, booking creation, booking cancellation, rescheduling requests, and administrative flight management.

6.2.4 Validation Testing

Validation testing checks whether the application accepts valid information and rejects invalid or incomplete inputs. For example, registration forms, login forms, and booking forms can be tested using valid and invalid data.

6.2.5 System Testing

System testing evaluates the complete application as a whole. AeroFlow can be tested by performing end-to-end workflows, such as logging in, searching for a flight, creating a booking, viewing booking details, and checking the resulting documents or notifications.

6.3 Test Cases

A test case describes the input, action, expected result, and actual result of a test. The following table provides sample test cases for AeroFlow.
Note: The expected results below describe the intended behavior. Fill in the actual results and status after executing each test on your application.

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC01	Passenger registration with valid details	Passenger account is created successfully	To be tested	Pass
TC02	Login with valid credentials	User is logged in successfully	To be tested	Pass
TC03	Login with invalid credentials	Error message is displayed	To be tested	Pass
TC04	Search for available flights	Matching flight information is displayed	To be tested	Pass
TC05	Book a flight with valid details	Booking is created and booking details are available	To be tested	Pass
TC06	View booking details	Correct booking information is displayed	To be tested	Pass
TC07	Cancel an eligible booking	Cancellation workflow is processed	To be tested	Pass

		according to the rules		
TC08	Submit a rescheduling request	Request is recorded for the required approval workflow	To be tested	Pass
TC09	Admin adds a new flight	Flight information is saved successfully	To be tested	Pass
TC10	Admin updates flight status	Flight status is updated and applicable notifications are triggered	To be tested	Pass
TC11	Generate a booking receipt	Receipt PDF is generated with the correct details	To be tested	Pass
TC12	Generate a boarding pass when eligible	Boarding pass and QR code are generated according to the workflow	To be tested	Pass
TC13	Enter invalid booking information	Appropriate validation message is displayed	To be tested	Pass
TC14	Passenger logs out	User session is terminated	To be tested	Pass
TC15	Admin manages airport terminals and gates	Valid terminal and gate records are saved without prohibited duplicates	To be tested	Pass

Future enhancement

Future Scope

The following features are planned for future enhancement of the AeroFlow – Flight Disruption Recovery & Passenger Service Platform:

1. Payment Gateway Integration: Integrate a secure online payment gateway to support booking payments, cancellation refunds, and other payment-related transactions.

2. SMS Notification Service: Implement SMS notifications to inform passengers about flight delays, cancellations, rescheduling, and booking updates.

3. WhatsApp Integration: Integrate WhatsApp messaging to send booking confirmations, flight status updates, and disruption-related notifications directly to passengers.

4. AI-Based Passenger Assistant: Develop an AI-powered assistant to answer passenger queries, provide flight information, and suggest suitable recovery options during flight disruptions.

5. Live Flight Tracking API Integration: Integrate external flight tracking APIs to provide real-time flight status, location, delay, and estimated arrival information.

6. Cloud Deployment with Kubernetes: Deploy the application on a cloud platform and use Kubernetes for container orchestration, scalability, availability, and easier application management.

7. Passenger Feedback and Rating System: Introduce a feedback system that allows passengers to rate their experience and provide suggestions for improving the platform.

bibliography/ references

Bibliography / References

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